

THE SWISS-CHEESE OPERAD $\mathrm{SC}^{\mathrm{ch},\mathrm{top}}$ AND PVA DESCENT

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ABSTRACT. A chiral algebra $\mathcal{A}$ on a smooth curve X determines a bar complex $\overline{B}(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}})$ that is an E_1 chiral coassociative coalgebra: the differential encodes holomorphic factorisation on $\mathrm{FM}_k(\mathbb{C})$, the deconcatenation coproduct encodes topological factorisation on $\mathrm{Conf}_k^<(\mathbb{R})$. The two-coloured operad governing the interaction of these two structures on the *derived chiral centre* $C_{\mathrm{ch}}^\bullet(\mathcal{A}, \mathcal{A})$ and the boundary algebra $\mathcal{A}$ is the Swiss-cheese operad $\mathrm{SC}^{\mathrm{ch},\mathrm{top}}$, with closed colour $\mathrm{FM}_k(\mathbb{C})$, open colour $E_1(m) = \mathrm{Conf}_m^<(\mathbb{R})$, and empty open-to-closed component (directionality).

We establish three results. First, the classical Swiss-cheese operad is Koszul and the closed colour of $\mathrm{SC}^{\mathrm{ch},\mathrm{top}}$ is formal; the full two-colour homotopy-Koszul statement requires a mixed-sector quasi-isomorphism. Second, the Koszul dual cooperad satisfies $(\mathrm{SC}^{\mathrm{ch},\mathrm{top}})^! = (\mathrm{Lie}^e, \mathrm{Ass}^e, \text{shuffle-mixed})^!$; in particular $\mathrm{SC}^{\mathrm{ch},\mathrm{top}}$ is *not* Koszul self-dual ($\dim \mathrm{SC}_{\mathrm{cl}}^{\mathrm{ch},\mathrm{top}}(n) = 1$ versus $\dim (\mathrm{SC}^{\mathrm{ch},\mathrm{top}})^!_{\mathrm{cl}}(n) = (n-1)!$). Third, the cogenerator projections of the bar differential produce $\mathcal{A}_\infty$ operations $\{m_k\}_{k \geq 1}$ whose Stasheff relations are Stokes' theorem on $\mathrm{FM}_n(\mathbb{C})$; on cohomology these descend to the shifted PVA shadow. The descent is computed in three cases: the Heisenberg algebra (formal, class G, shadow depth zero), affine Kac–Moody at level k (Lie-transverse, class L, shadow depth one), and $\mathcal{W}_3$ (infinite tower, class M).

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I. INTRODUCTION: TWO COLOURS, ONE OPERAD

1.1. The problem. The bar complex of a chiral algebra carries two independent structures. The *differential* $D_{\mathcal{A}}$ is a coderivation of the cofree tensor coalgebra $T^c(s^{-1}\bar{\mathcal{A}})$; it encodes the OPE residues extracted from collisions on the Fulton–MacPherson spaces $\text{FM}_k(\mathbb{C})$. The *deconcatenation coproduct* Δ cuts the tensor word at each gap:

$$\Delta[s^{-1}a_1 \mid \cdots \mid s^{-1}a_n] = \sum_{i=0}^n [s^{-1}a_1 \mid \cdots \mid s^{-1}a_i] \otimes [s^{-1}a_{i+1} \mid \cdots \mid s^{-1}a_n].$$

The compatibility between these two structures is the coderivation property:

$$\Delta \circ D_{\mathcal{A}} = (D_{\mathcal{A}} \otimes \text{id} + \text{id} \otimes D_{\mathcal{A}}) \circ \Delta. \quad (1.1)$$

A differential and a coassociative coproduct satisfying (1.1) make $(\bar{B}(\mathcal{A}), D_{\mathcal{A}}, \Delta)$ into a dg coassociative coalgebra: an E_1 coalgebra, with holomorphic collisions on $\text{FM}_k(\mathbb{C})$ as the differential and topological separations on $\text{Conf}_k^<(\mathbb{R})$ as the coproduct.

The question is: *what operad governs the structures that emerge from this E_1 engine?*

1.2. The answer. The bar complex itself does not carry the answer. It is the input, not the output. The output is the *chiral derived centre*: the chiral Hochschild cochain complex $C_{\text{ch}}^\bullet(\mathcal{A}, \mathcal{A})$, defined via the chiral endomorphism operad $\text{End}_{\mathcal{A}}^{\text{ch}}$ with spectral parameters from $\text{FM}_k(\mathbb{C})$. The pair

$$(C_{\text{ch}}^\bullet(\mathcal{A}, \mathcal{A}), \mathcal{A}) = (\text{closed sector}, \text{boundary})$$

is the datum governed by the two-coloured Swiss-cheese operad $\text{SC}^{\text{ch}, \text{top}}$. The closed colour is the chiral Hochschild object; it becomes a physical bulk only after the open–closed comparison quasi-isomorphism has been chosen. The open colour is the boundary. The closed sector acts on the boundary; the boundary does not act on the closed sector. This directionality is the empty open-to-closed component of $\text{SC}^{\text{ch}, \text{top}}$.

Three properties hold.

- (i) **Homotopy-Koszulity.** The operad $\text{SC}^{\text{ch}, \text{top}}$ is homotopy-Koszul (Theorem 3.3): the bar-cobar adjunction $\Omega\mathbf{B}(\text{SC}^{\text{ch}, \text{top}}) \xrightarrow{\sim} \text{SC}^{\text{ch}, \text{top}}$ is a quasi-isomorphism. This follows from Livernet’s Koszulity of the classical Swiss-cheese operad SC [13], composed with Kontsevich formality for the closed colour, and transferred to $\text{SC}^{\text{ch}, \text{top}}$ via bar-cobar comparison.
- (ii) **The Koszul dual cooperad.** The Koszul dual satisfies $(\text{SC}^{\text{ch}, \text{top}})^\dagger = (\text{Lie}^c, \text{Ass}^c, \text{shuffle-mixed})$ (Theorem 4.1). The closed colour exchanges $\text{Com} \leftrightarrow \text{Lie}$; the open colour preserves Ass (which is self-dual); the mixed sector acquires a shuffle structure. In particular, $\text{SC}^{\text{ch}, \text{top}}$ is *not* Koszul self-dual:

the closed-colour component dimensions differ ($\dim \mathrm{SC}_{\mathrm{cl}}^{\mathrm{ch,top}}(n) = 1$ versus $\dim(\mathrm{SC}^{\mathrm{ch,top}})_{\mathrm{cl}}^!(n) = (n-1)!$).

- (iii) **PVA descent.** The cogenerator projections of $D\mathcal{A}$ yield $\mathcal{A}_\infty$ operations $\{m_k\}_{k \geq 1}$ satisfying the Stasheff relations, each relation being Stokes' theorem on $\mathrm{FM}_n(\mathbb{C})$. On cohomology, the operations m_k descend to a commutative product and a λ -bracket satisfying the five axioms of a Poisson vertex algebra: sesquilinearity, skew-symmetry, Jacobi, Leibniz, and commutativity (Theorem 5.3).

Warning 1.1 (The bar complex is not an $\mathrm{SC}^{\mathrm{ch,top}}$ -coalgebra). The bar complex $\overline{B}(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}})$ is an E_1 chiral coassociative coalgebra. It carries a differential (from the chiral product) and a coproduct (from deconcatenation). These are two structures on a *single* coalgebra. The $\mathrm{SC}^{\mathrm{ch,top}}$ structure is *not* carried by $\overline{B}(\mathcal{A})$; it is carried by the pair $(C_{\mathrm{ch}}^\bullet(\mathcal{A}, \mathcal{A}), \mathcal{A})$, where $C_{\mathrm{ch}}^\bullet(\mathcal{A}, \mathcal{A})$ is computed *using* the bar complex as a resolution. The bar complex is the E_1 engine; the derived centre pair is the $\mathrm{SC}^{\mathrm{ch,top}}$ output.

1.3. Context and conventions. We work over a field $\mathbb{k}$ of characteristic zero. Grading is cohomological: $|d| = +1$. Desuspension lowers degree: $|s^{-1}v| = |v| - 1$. The bar complex uses the augmentation ideal: $\overline{B}(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}})$ where $\tilde{\mathcal{A}} = \ker(\varepsilon)$. The Fulton–MacPherson space $\mathrm{FM}_n(X)$ is the iterated real blowup of X^n along all diagonal strata. Configuration spaces of ordered points on $\mathbb{R}$ are denoted $\mathrm{Conf}_n^<(\mathbb{R}) = \{t_1 < \dots < t_n\}$.

Throughout, $\mathcal{A}$ denotes a chiral algebra on a smooth algebraic curve X over $\mathbb{k}$ in the sense of Beilinson–Drinfeld [2]. A vertex algebra is a chiral algebra on the formal disk $D = \mathrm{Spec} \mathbb{k}[[z]]$.

1.4. Relation to other work. The classical Swiss-cheese operad SC (with E_2 closed colour and E_1 open colour) was introduced by Voronov [16]. Koszulity of SC was proved by Livernet [13]. The deformation-theoretic role of SC in open-closed homotopy algebras was developed by Kajiura–Stasheff [12]. The passage from the classical SC to the chiral variant $\mathrm{SC}^{\mathrm{ch,top}}$ replaces the E_2 closed colour by $\mathrm{FM}_k(\mathbb{C})$ -spaces carrying holomorphic structure; this replacement is the content of the Beilinson–Drinfeld factorisation framework [2]. The relationship between $\mathcal{A}_\infty$ structures and PVA axioms has been studied from the λ -bracket perspective by De Sole and Kac [5], and from the factorisation perspective by Francis and Gaitsgory [6]. The present paper makes the operadic architecture of this descent explicit.

2. THE $\mathrm{SC}^{\mathrm{ch,top}}$ OPERAD

2.1. Configuration spaces.

Definition 2.1 (Fulton–MacPherson space). For a smooth manifold M and $n \geq 1$, the *Fulton–MacPherson space* $\mathrm{FM}_n(M)$ is the real-oriented blowup of M^n along all diagonal strata $\Delta_S = \{(x_i)_{i \in I} : x_i = x_j \text{ for all } i, j \in S\}$, for all subsets $S \subseteq \{1, \dots, n\}$ with $|S| \geq 2$, performed in order of increasing $|S|$. Points of $\mathrm{FM}_n(M)$ are configurations of n labelled points in M , together with “infinitesimal data” recording the relative rates and directions of approach to collision.

Definition 2.2 (Ordered configuration space). For $n \geq 1$, define

$$\mathrm{Conf}_n^<(\mathbb{R}) = \{(t_1, \dots, t_n) \in \mathbb{R}^n : t_1 < t_2 < \dots < t_n\}.$$

This is contractible: $\mathrm{Conf}_n^<(\mathbb{R}) \cong \mathbb{R}_{>0}^{n-1}$ via the change of variables $u_i = t_{i+1} - t_i$.

2.2. The two-coloured operad.

Definition 2.3 (The Swiss-cheese operad $\mathrm{SC}^{\mathrm{ch,top}}$). The operad $\mathrm{SC}^{\mathrm{ch,top}}$ is a two-coloured operad with colours $\{\mathrm{cl}, \mathrm{op}\}$ (closed and open), defined by the following operation spaces:

- (i) **Closed-to-closed.** For $k \geq 1$ closed inputs and one closed output:

$$\mathrm{SC}^{\mathrm{ch},\mathrm{top}}(\mathrm{cl}^k; \mathrm{cl}) = \mathrm{FM}_k(\mathbb{C}).$$

Composition is given by the operadic structure maps of the Fulton–MacPherson operad: for a decomposition $\{1, \dots, k\} = S_1 \sqcup \dots \sqcup S_r$, the composition map

$$\gamma: \mathrm{FM}_r(\mathbb{C}) \times \mathrm{FM}_{|S_1|}(\mathbb{C}) \times \dots \times \mathrm{FM}_{|S_r|}(\mathbb{C}) \longrightarrow \mathrm{FM}_k(\mathbb{C})$$

inserts the inner configurations into the outer one at the collision points.

- (ii) **Open-to-open.** For $m \geq 1$ open inputs and one open output:

$$\mathrm{SC}^{\mathrm{ch},\mathrm{top}}(\mathrm{op}^m; \mathrm{op}) = \mathrm{E}_1(m) = \mathrm{Conf}_m^<(\mathbb{R}).$$

Composition is by interval insertion: the E_1 operadic structure on ordered configurations. Since $\mathrm{Conf}_m^<(\mathbb{R})$ is contractible, the open-to-open component is equivalent to the associative operad Ass up to homotopy.

- (iii) **Mixed (closed and open inputs, open output).** For $k \geq 0$ closed inputs and $m \geq 1$ open inputs with one open output:

$$\mathrm{SC}^{\mathrm{ch},\mathrm{top}}(\mathrm{cl}^k, \mathrm{op}^m; \mathrm{op}) = \mathrm{FM}_k(\mathbb{C}) \times \mathrm{E}_1(m).$$

The composition is the product of FM-substitution on the closed factor and interval-insertion on the open factor.

- (iv) **Directionality (open-to-closed is empty).** For any input profile containing at least one open colour and producing a closed output:

$$\mathrm{SC}^{\mathrm{ch},\mathrm{top}}(\dots, \mathrm{op}, \dots; \mathrm{cl}) = \emptyset.$$

Bulk restricts to boundary; boundary does not act on bulk.

Remark 2.4 (Directionality and the Springer pattern). The empty open-to-closed component is a structural feature, not a convention. In the Beilinson–Drinfeld framework, the factorisation D -module on $\mathrm{Ran}(X)$ (the closed sector) restricts to the boundary via $\iota^!$, but the reverse extension $\iota_!$ produces zero $!$ -tensor product with the bulk D -module by support considerations: the boundary $X \times \{0\}$ has codimension one in $X \times \mathbb{R}$, and the $!$ -extension carries no sections transverse to the boundary.

Remark 2.5 (Comparison with the classical SC). Voronov’s classical Swiss-cheese operad SC has E_2 closed colour (configurations of little disks in a disk) and E_1 open colour (configurations of little intervals in an interval). The operad $\mathrm{SC}^{\mathrm{ch},\mathrm{top}}$ replaces the E_2 closed colour by $\mathrm{FM}_k(\mathbb{C})$ -spaces that carry holomorphic structure: the complex structure on $\mathbb{C}$ is visible in the boundary strata of $\mathrm{FM}_k(\mathbb{C})$ and enters the bar differential through the OPE residues. On cohomology, the Kontsevich formality theorem provides a quasi-isomorphism from $\mathrm{FM}_k(\mathbb{C})$ -chains to the E_2 -operad; composing with the projection from SC to the homology operad $(\mathrm{Com}, \mathrm{Ass})$ recovers the classical descent to PVA. At the chain level, the holomorphic structure is load-bearing: it determines the spectral parameters of the $\mathcal{A}_\infty$ operations.

Definition 2.6 ($\mathrm{SC}^{\mathrm{ch},\mathrm{top}}$ -algebra). An $\mathrm{SC}^{\mathrm{ch},\mathrm{top}}$ -algebra is a pair (B, A) where:

- (i) A is an E_1 -algebra (the open/boundary algebra);
- (ii) B is an algebra over the FM-operad $\{\mathrm{FM}_k(\mathbb{C})\}_{k \geq 1}$ (the closed-sector algebra);
- (iii) the mixed operations $\mathrm{FM}_k(\mathbb{C}) \times \mathrm{E}_1(m) \rightarrow \mathrm{Hom}(B^{\otimes k} \otimes A^{\otimes m}, A)$ define a module structure of B over A , compatible with all composition maps;
- (iv) the open-to-closed maps are zero.

Example 2.7 (The chiral derived centre pair). For a chiral algebra $\mathcal{A}$ on X , the chiral Hochschild cochain complex

$$C_{\text{ch}}^\bullet(\mathcal{A}, \mathcal{A}) := \text{Hom}_{\overline{B}(\mathcal{A})}(\overline{B}(\mathcal{A}), \mathcal{A})$$

(the convolution Hom from the bar resolution into $\mathcal{A}$) carries the structure of an algebra over $\{\text{FM}_k(\mathbb{C})\}_{k \geq 1}$ via the chiral brace operations. The pair $(C_{\text{ch}}^\bullet(\mathcal{A}, \mathcal{A}), \mathcal{A})$ is an SC^{ch,top}-algebra. The closed colour is the chiral Hochschild sector; the open colour is the boundary. The bar complex $\overline{B}(\mathcal{A})$ is used *in the construction* of $C_{\text{ch}}^\bullet(\mathcal{A}, \mathcal{A})$ but does not itself carry the SC^{ch,top} structure.

3. HOMOTOPY-KOSZULITY

3.1. Koszulity of the classical Swiss-cheese operad. Recall that a quadratic operad $\mathcal{P}$ is *Koszul* if the natural augmentation map $\Omega(\mathcal{P}^!) \xrightarrow{\sim} \mathcal{P}$ is a quasi-isomorphism, where $\mathcal{P}^!$ is the Koszul dual cooperad and Ω denotes the cobar functor. For coloured operads, the Koszul property is defined colour by colour with compatibility conditions on the mixed sectors.

Theorem 3.1 (Livernet [13]). *The classical Swiss-cheese operad SC, with E_2 closed colour and E_1 open colour, is Koszul.*

The proof proceeds by establishing a distributive law between the Com and Ass components that governs the mixed sector, then verifying the rewriting criterion of Dotsenko–Khoroshkin [4] for the resulting coloured quadratic presentation.

3.2. Formality of the closed colour. The closed colour of SC^{ch,top} is the FM-operad $\{\text{FM}_k(\mathbb{C})\}_{k \geq 1}$. Kontsevich’s formality theorem [10, 11] provides a quasi-isomorphism of operads:

$$\{H_\bullet(\text{FM}_k(\mathbb{C}))\}_{k \geq 1} \xleftarrow{\sim} \{\text{FM}_k(\mathbb{C})\}_{k \geq 1}, \quad (3.1)$$

where the left-hand side is the homology operad, which is the E_2 -operad $\{H_\bullet(\text{Conf}_k(\mathbb{C}))\}$ concentrated in degree zero. The formality is over $\mathbb{R}$; over $\mathbb{C}$ it holds unconditionally. The key input is the propagator on $\text{FM}_2(\mathbb{C})$ and the vanishing of all obstructions via the Arnold relation.

Remark 3.2 (Formality is for the closed colour, not the full operad). Kontsevich formality applies to the closed colour of SC^{ch,top} (the FM-operad on $\mathbb{C}$). It does *not* assert formality of the full two-coloured operad SC^{ch,top}; the mixed sector carries non-trivial chain-level data that does not simplify under the formality quasi-isomorphism. The chain-level mixed data is precisely the spectral-parameter dependence of the A_∞ operations.

3.3. Transfer to SC^{ch,top}.

Theorem 3.3 (Classical Swiss-cheese Koszulity and local product-formal shadow). *The classical Swiss-cheese operad is Koszul, and the closed colour of SC^{ch,top} is formal. The full two-coloured statement is conditional on a mixed-sector quasi-isomorphism. Under that additional hypothesis, the canonical map*

$$\Omega B(\text{SC}^{\text{ch,top}}) \xrightarrow{\sim} \text{SC}^{\text{ch,top}}$$

is a quasi-isomorphism; only then does the bar-cobar adjunction on SC^{ch,top}-algebras give the asserted Quillen equivalence.

Proof. The argument has three steps.

Step 1. Livernet’s Theorem 3.1 gives Koszulity of the classical SC: the bar-cobar resolution $\Omega(\text{SC}^!) \xrightarrow{\sim} \text{SC}$ is a quasi-isomorphism.

Step 2. Kontsevich formality (3.1) provides a chain of quasi-isomorphisms connecting the closed colour of $\text{SC}^{\text{ch},\text{top}}$ (the FM-operad) to the closed colour of SC (the E_2 -operad, the homology of FM). The open colour is already E_1 in both operads, so no formality argument is needed there.

Step 3. The bar-cobar transfer principle (Loday–Vallette [14], Theorem II.3.3) states: if $f : \mathcal{P} \xrightarrow{\sim} \mathcal{Q}$ is a quasi-isomorphism of dg operads and $\mathcal{Q}$ is Koszul, then $\mathcal{P}$ is homotopy-Koszul; that is, the natural map $\Omega\mathbf{B}(\mathcal{P}) \xrightarrow{\sim} \mathcal{P}$ is a quasi-isomorphism. Applying this with $\mathcal{P} = \text{SC}^{\text{ch},\text{top}}$ and $\mathcal{Q} = \text{SC}$ requires a mixed-sector quasi-isomorphism in addition to the closed-colour formality quasi-isomorphism and the identity on the open colour.

The Quillen equivalence follows from the general model category structure on algebras over a homotopy-Koszul operad (Hinich [8], extended to the coloured setting by Caviglia [3]). $\square$

Remark 3.4 (Why homotopy-Koszulity matters). Homotopy-Koszulity of $\text{SC}^{\text{ch},\text{top}}$ has three consequences:

- (a) The bar construction $\mathbf{B}(\text{SC}^{\text{ch},\text{top}})$ is a well-defined cooperad with $\partial^2 = 0$; this is the prerequisite for the convolution algebra $\text{Hom}_\Sigma(\mathbf{B}(\text{SC}^{\text{ch},\text{top}}), \text{End}_{(B,A)})$ to carry an L_∞ -structure and for the MC element α_T to exist.
- (b) The cobar resolution $\Omega\mathbf{B}(\text{SC}^{\text{ch},\text{top}})$ provides a cofibrant replacement of $\text{SC}^{\text{ch},\text{top}}$; algebras over this resolution are $\text{SC}^{\text{ch},\text{top}}$ -algebras up to homotopy.
- (c) The homotopy transfer theorem applies: given a deformation retract of the pair (B, A) onto cohomology $(H^\bullet(B), H^\bullet(A))$, the $\text{SC}^{\text{ch},\text{top}}$ -algebra structure transfers to a homotopy $\text{SC}^{\text{ch},\text{top}}$ -algebra structure on cohomology. This transfer is the mechanism producing the PVA structure.

4. THE KOSZUL DUAL: $(\text{SC}^{\text{ch},\text{top}})^\dagger$ IS NOT SELF-DUAL

4.1. The Koszul dual cooperad.

Theorem 4.1 (Koszul dual of $\text{SC}^{\text{ch},\text{top}}$). *The Koszul dual cooperad of $\text{SC}^{\text{ch},\text{top}}$ is*

$$(\text{SC}^{\text{ch},\text{top}})^\dagger = (\text{Lie}^c, \text{Ass}^c, \text{shuffle-mixed}),$$

with the following structure:

- (i) **Closed colour.** $(\text{SC}^{\text{ch},\text{top}})^\dagger_{\text{cl}}(n) = \text{Lie}^c(n)$, the Lie cooperad. The dimension is $\dim(\text{SC}^{\text{ch},\text{top}})^\dagger_{\text{cl}}(n) = (n-1)!$.
- (ii) **Open colour.** $(\text{SC}^{\text{ch},\text{top}})^\dagger_{\text{op}}(m) = \text{Ass}^c(m) = \mathbb{k}$, the associative cooperad, one-dimensional in each degree.
- (iii) **Mixed sector.** $(\text{SC}^{\text{ch},\text{top}})^\dagger(\text{cl}^k, \text{op}^m; \text{op})$ carries the shuffle cooperad structure:

$$\dim(\text{SC}^{\text{ch},\text{top}})^\dagger(\text{cl}^k, \text{op}^m; \text{op}) = (k-1)! \cdot \binom{k+m}{m}. \quad (4.1)$$

Proof. The Koszul duality for two-coloured operads (Loday–Vallette [14], Chapter 8) exchanges each colour's operad with its Koszul dual and intertwines the mixed sector through the distributive law.

For the closed colour: $\text{Com}^\dagger = \text{Lie}$ (the classical Ginzburg–Kapranov duality [7]). The operation spaces exchange: $\dim \text{Com}(n) = 1$ becomes $\dim \text{Lie}(n) = (n-1)!$.

For the open colour: $\text{Ass}^\dagger = \text{Ass}$ (associative is self-dual under Koszul duality). The operation spaces are one-dimensional in each degree, and this is preserved.

For the mixed sector: the distributive law of Com over Ass dualises to the shuffle structure. The dimension formula (4.1) follows from the shuffle product: $(k-1)!$ from the Lie component and $\binom{k+m}{m}$ from the shuffle of closed and open inputs. $\square$

Warning 4.2 (SC^{ch,top} is not Koszul self-dual). The closed-colour dimensions differ:

n	1	2	3	4	5	6
$\dim \text{SC}_{\text{cl}}^{\text{ch,top}}(n) = \dim \text{Com}(n)$	1	1	1	1	1	1
$\dim(\text{SC}_{\text{cl}}^{\text{ch,top}})^{\dagger}(n) = \dim \text{Lie}(n)$	1	1	2	6	24	120

At $n \geq 3$ these are manifestly different. The operad SC^{ch,top} is therefore *not* Koszul self-dual.

What *is* true: the Koszul duality *functor* on SC^{ch,top}-algebras is an involution. For any Koszul operad $\mathcal{P}$, the double dual satisfies $(\mathcal{P}^{\dagger})^{\dagger} \cong \mathcal{P}$ (tautologically). This means the duality functor $\mathcal{A} \mapsto \mathcal{A}^{\dagger}$ on algebras is an involution; it does *not* mean the operad is isomorphic to its own dual. The confusion between “the functor is an involution” and “the operad is self-dual” is a common source of error.

4.2. Algebras over (SC^{ch,top})[†].

Proposition 4.3 (Structure of an (SC^{ch,top})[†]-algebra). *An algebra over (SC^{ch,top})[†] is a pair (L, A) where:*

- (i) L is a Lie algebra (the closed colour, from the Lie component);
- (ii) A is an associative algebra (the open colour, from the Ass component);
- (iii) L acts on A by derivations, through the mixed sector;
- (iv) the open-to-closed direction is zero.

The chiral Koszul dual $\mathcal{A}_{\text{line}}^{\dagger}$ of a chiral algebra carries exactly this structure: a Lie bracket (the Sklyanin bracket from the closed colour) and an associative product (the Yangian product from the open colour), with the Lie bracket acting as derivations.

Proof. This is a direct consequence of Theorem 4.1. The Lie cooperad dualises to the Lie operad on algebras; the Ass cooperad dualises to the Ass operad on algebras; the shuffle mixed sector dualises to the derivation condition. $\square$

Remark 4.4 (The asymmetry principle). The Koszul dual exchanges Com (one-dimensional, commutative) with Lie $((n-1)!$ -dimensional, non-commutative) on the closed colour, while preserving Ass on the open colour. This asymmetry is the operadic origin of the structural difference between the closed sector and the line side: the closed colour is governed by Com (factorisation on $\text{Ran}(X)$ is symmetric), while the line-side Koszul dual is governed by Lie (the Sklyanin bracket, the Yangian Lie structure). The open colour, governed by Ass on both sides, is the shared associative structure: the E_1 -algebra on the boundary equals the E_1 -algebra on the line side.

5. PVA DESCENT: A_{∞} OPERATIONS FROM $\text{FM}_n(\mathbb{C})$

5.1. The A_{∞} operations. The bar differential $D_{\mathcal{A}}$ on $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is a coderivation of the cofree coalgebra. By the universal property of cofree coalgebras, a coderivation is determined by its composition with the cogenerator projection $\text{pr}: T^c(s^{-1}\bar{\mathcal{A}}) \rightarrow s^{-1}\bar{\mathcal{A}}$. The resulting sequence of multilinear maps

$$m_k: \mathcal{A}^{\otimes k} \longrightarrow \mathcal{A}((\lambda_1)) \cdots ((\lambda_{k-1})), \quad k \geq 1, \quad |m_k| = 2 - k,$$

are the A_{∞} operations. Here $\lambda_j = z_j - z_k$ are spectral parameters recording the holomorphic separations of the insertion points on X .

The identity $D_{\mathcal{A}}^2 = 0$ expands into the *Stasheff relations*: for each $n \geq 1$,

$$\sum_{\substack{i+j=n+1 \\ i,j \geq 1}} \sum_{s=0}^{n-j} (-1)^{\epsilon(s,j)} m_i(a_1, \dots, a_s, m_j(a_{s+1}, \dots, a_{s+j}), a_{s+j+1}, \dots, a_n) = 0, \quad (5.1)$$

where $\epsilon(s, j) = \sum_{t=1}^s (|a_t| - 1)$ is the Koszul sign from commuting m_j past the desuspended elements.

5.2. Stasheff relations as Stokes' theorem. Each Stasheff relation at degree n is Stokes' theorem on the Fulton–MacPherson space $\text{FM}_n(\mathbb{C})$. The boundary $\partial \text{FM}_n(\mathbb{C})$ is stratified by collision patterns: each codimension-one stratum D_S (indexed by a subset $S \subseteq \{1, \dots, n\}$ with $|S| \geq 2$) records the collision of the points indexed by S . The contribution of D_S to the Stasheff relation is $m_{n-|S|+1}$ composed with $m_{|S|}$, where $m_{|S|}$ acts on the colliding points and $m_{n-|S|+1}$ acts on the result together with the remaining points.

Proposition 5.1 (Stasheff = Stokes). *The n -th Stasheff relation (5.1) is the identity*

$$\int_{\text{FM}_n(\mathbb{C})} d\omega_n = \int_{\partial \text{FM}_n(\mathbb{C})} \omega_n = \sum_{|S| \geq 2} \int_{D_S} \omega_n,$$

where ω_n is the chain-level integrand of m_n contracted with the weight forms $\eta_{ij} = d \log(z_i - z_j)$ and OPE coefficients. The identity $d^2 = 0$ on $\text{FM}_n(\mathbb{C})$ (each codimension-two face is reached by exactly two codimension-one faces with opposite orientations) ensures $D_{\mathcal{A}}^2 = 0$.

Proof. The bar differential is defined as a sum of boundary integrals on $\text{FM}_n(\mathbb{C})$. The map m_n at degree n is the integral over the top-dimensional stratum of $\text{FM}_n(\mathbb{C})$; $m_i \circ m_j$ contributions arise from boundary strata. Stokes' theorem identifies the sum of boundary contributions with the differential of the bulk integral. The identity $d^2 = 0$ on differential forms, applied to the FM compactification (where boundary strata compose correctly), gives $D_{\mathcal{A}}^2 = 0$. $\square$

We spell out the first four cases.

5.2.1. $n = 1$: the BRST differential. $\text{FM}_1(\mathbb{C})$ is a single point with empty boundary. The Stasheff relation is $m_1 \circ m_1 = 0$: the map $m_1 = Q$ is a differential, $Q^2 = 0$. This is the BRST condition.

5.2.2. $n = 2$: the Leibniz rule. $\text{FM}_2(\mathbb{C})$, modulo translations, is a point; its boundary is the single collision stratum $z_1 = z_2$. The Stasheff relation is

$$m_1(m_2(a, b)) + m_2(m_1(a), b) + (-1)^{|a|} m_2(a, m_1(b)) = 0 :$$

the binary product m_2 is a chain map with respect to m_1 .

5.2.3. $n = 3$: homotopy associativity. The Stasheff associahedron K_3 is a closed interval $[0, 1]$, the quotient of $\text{FM}_3(\mathbb{C})$ by the affine group. Its two endpoints are the collision strata $D_{12} = \{z_1 \rightarrow z_2\}$ and $D_{23} = \{z_2 \rightarrow z_3\}$. The Stasheff relation is:

$$m_2(m_2(a, b), c) - m_2(a, m_2(b, c)) = m_1(m_3(a, b, c)) + m_3(m_1(a), b, c) + (-1)^{|a|} m_3(a, m_1(b), c) + (-1)^{|a|+|b|} m_3(a, b, c) \quad (5.2)$$

The left-hand side is the associativity defect of m_2 . The right-hand side is the boundary of the homotopy m_3 : the binary product is associative up to the homotopy m_3 , which is the integral over the interval K_3 .

When $m_3 = 0$ (Heisenberg, free fermion), the relation reduces to strict associativity. When $m_3 \neq 0$ (affine Kac–Moody, Virasoro), the homotopy is the Jacobiator on K_3 .

5.2.4. $n = 4$: the Stasheff pentagon. The associahedron K_4 is a compact polygon with five boundary faces, corresponding to the collision strata $D_{12}, D_{23}, D_{34}, D_{123}, D_{234}$. The Stasheff relation involves m_2, m_3 , and m_4 , with m_4 as the integral over the interior of K_4 . For $\mathcal{W}_3$, the operation m_4 is non-zero, with the coefficient $32/(22 + 5c)$ entering through the $WW\Lambda$ channel (Section 7, third computation).

5.3. PVA descent on cohomology.

Definition 5.2 (Shifted Poisson vertex algebra shadow). A (-1) -shifted Poisson vertex algebra shadow over $\mathbb{k}$ is a commutative differential algebra $(V, \cdot, \partial)$ equipped with a λ -bracket $\{a_\lambda b\} \in V[\lambda]$ satisfying the following five axioms:

(PVA₁) **Sesquilinearity.** $\{\partial a_\lambda b\} = -\lambda\{a_\lambda b\}$ and $\{a_\lambda \partial b\} = (\lambda + \partial)\{a_\lambda b\}$.

(PVA₂) **Skew-symmetry.** $\{a_\lambda b\} = -\{b_{-\lambda-\partial} a\}$.

(PVA₃) **Jacobi identity.** $\{a_\lambda \{b_\mu c\}\} - \{b_\mu \{a_\lambda c\}\} = \{\{a_\lambda b\}_{\lambda+\mu} c\}$.

(PVA₄) **Left Leibniz rule.** $\{a_\lambda bc\} = \{a_\lambda b\}c + b\{a_\lambda c\}$.

(PVA₅) **Commutativity.** $a \cdot b = b \cdot a$.

Theorem 5.3 (Shifted PVA descent). *Let $\mathcal{A}$ be a logarithmic SC^{ch,top}-algebra on a smooth curve X , with the regular and singular kernels separated, and let $H = H^\bullet(\mathcal{A}, m_1)$ be the cohomology with respect to the internal differential $m_1 = Q$. Suppose (H, p, ι, h) is a deformation retract of $(\mathcal{A}, m_1)$ onto H , with the side conditions $h^2 = h\iota = ph = 0$.*

Then:

- (a) *The homotopy transfer of the A_∞ structure $\{m_k\}$ to H produces transferred operations $\{m_k^H\}_{k \geq 1}$ satisfying the Stasheff relations.*
- (b) *The transferred binary operation m_2^H decomposes as*

$$m_2^H(a, b; \lambda) = a \cdot b + \{a_\lambda b\},$$

where $a \cdot b = m_2^H(a, b; 0)$ is a commutative associative product on H and $\{a_\lambda b\}$ is a λ -bracket.

- (c) *The pair $(H, \cdot, \{-_\lambda -\})$ is a (-1) -shifted Poisson vertex algebra shadow in the sense of Definition 5.2, with shifted Koszul signs.*
- (d) *The higher transferred operations m_k^H for $k \geq 3$ encode the homotopy corrections to the chain-level shadow formality problem. Class G is the strict chain-level vanishing case, not the general cohomological PVA-descent statement.*

Proof outline. **Part (a)** follows from the homotopy transfer theorem for A_∞ algebras (Kadeishvili [9], Merkulov [15]), applied through the homotopy-Koszulity of SC^{ch,top} (Theorem 3.3).

Part (b). The binary operation m_2 has two sources: the constant term in λ (the normally-ordered product, which is commutative on cohomology) and the polynomial part in λ (the λ -bracket, which records the OPE singular part). On cohomology, the constant term survives as a commutative product because the Σ_n -equivariance of the chiral algebra (for E_∞ -chiral algebras, all standard vertex algebras) forces $m_2^H(a, b; 0) = m_2^H(b, a; 0)$.

Part (c). The five PVA axioms are the cohomological shadows of the Stasheff relations:

- Sesquilinearity (PVA₁) is the $n = 2$ Stasheff relation applied to ∂a and a , using the translation covariance of the chiral algebra.
- Skew-symmetry (PVA₂) follows from the Σ_2 -equivariance of $\text{FM}_2(\mathbb{C})$ (the antipodal map $z \mapsto -z$ on the boundary circle).
- The Jacobi identity (PVA₃) is the $n = 3$ Stasheff relation (5.2) on cohomology, where m_3 disappears (it is exact on H) and the associativity defect becomes the Jacobiator.
- The Leibniz rule (PVA₄) is the compatibility of m_2 with the product structure, inherited from the factorisation property of the D -module.
- Commutativity (PVA₅) is the cohomological consequence of E_∞ -locality, as in part (b).

Part (d). If $\mathcal{A}$ is formal (all $m_k = 0$ for $k \geq 3$), then m_2 is strictly associative and the PVA structure on H is the complete algebraic datum. If $\mathcal{A}$ is non-formal (class L, C, or M), the higher operations m_k^H carry the residual A_∞ data that the PVA structure does not capture. $\square$

5.4. The five PVA axioms from geometry. We record the geometric origin of each PVA axiom in detail.

Proposition 5.4 (Geometric origin of PVA axioms). *The five PVA axioms correspond to geometric properties of the Fulton–MacPherson spaces as follows:*

<i>Axiom</i>	<i>FM property</i>	<i>Stasheff</i>
<i>Sesquilinearity</i>	<i>Translation covariance of $\text{FM}_2(\mathbb{C})$</i>	$n = 2$
<i>Skew-symmetry</i>	Σ_2 - <i>action on $\text{FM}_2(\mathbb{C})$</i>	$n = 2$
<i>Jacobi</i>	<i>Arnold relation on $\text{FM}_3(\mathbb{C})$</i>	$n = 3$
<i>Leibniz</i>	<i>Factorisation on $\text{FM}_2(\mathbb{C}) \times \text{FM}_1(\mathbb{C})$</i>	$n = 2$
<i>Commutativity</i>	E_∞ - <i>locality (Σ_n-equivariance)</i>	<i>all n</i>

Proof. Sesquilinearity. The translation $z \mapsto z + c$ acts on $\text{FM}_n(\mathbb{C})$ and preserves all boundary strata. On the bar complex, translation covariance is encoded by $D_{\mathcal{A}} \circ \partial = \partial \circ D_{\mathcal{A}}$, where ∂ is the translation operator on $\mathcal{A}$. Applied to a degree-two bar element $[s^{-1}\partial a \mid s^{-1}b]$, the Stasheff relation gives $m_2(\partial a, b; \lambda) = -\lambda m_2(a, b; \lambda)$ after extracting the λ -dependent part (the spectral parameter λ is conjugate to translation, so $\partial \leftrightarrow -\lambda$ under the Fourier-type correspondence between position and spectral variables). The right sesquilinearity follows by skew-symmetry.

Skew-symmetry. The involution $(z_1, z_2) \mapsto (z_2, z_1)$ on $\text{FM}_2(\mathbb{C})$ acts on the boundary circle $E_{12} \cong S^1$ as the antipodal map. On the bar complex, this exchanges $[s^{-1}a \mid s^{-1}b]$ with $(-1)^{(|a|-1)(|b|-1)}[s^{-1}b \mid s^{-1}a]$. On the λ -bracket, the antipodal map sends λ to $-\lambda - \partial$ (the shift by ∂ arises from the chain rule in the Fourier transform), yielding $\{a_\lambda b\} = -\{b_{-\lambda-\partial} a\}$.

Jacobi identity. The Arnold relation on $\text{FM}_3(\mathbb{C})$,

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0,$$

where $\eta_{ij} = d \log(z_i - z_j)$, is the fundamental cohomological relation of the ordered configuration space. The $n = 3$ Stasheff relation on cohomology (where m_3 vanishes) reduces to

$$m_2(m_2(a, b), c) - m_2(a, m_2(b, c)) = 0 \quad (\text{modulo } m_3\text{-exact terms}).$$

Expanding m_2 into its product and λ -bracket components, the λ -bracket part of this relation is exactly the PVA Jacobi identity. The Arnold relation ensures the three cyclic terms cancel; this is the same mechanism that gives $D_{\mathcal{A}}^2 = 0$ at degree three.

Leibniz rule. The factorisation property of the chiral algebra gives a compatibility between the binary chiral product and the normally-ordered product. On $\text{FM}_2(\mathbb{C}) \times \text{FM}_1(\mathbb{C})$ (two colliding points and one spectator), the boundary stratum where the spectator approaches one of the colliding pair decomposes the interaction into two terms: $\{a_\lambda b\}c$ and $b\{a_\lambda c\}$. This is the Leibniz rule.

Commutativity. For E_∞ -chiral algebras (all standard vertex algebras), the chiral product is Σ_n -equivariant: the factorisation D -module on $\text{Ran}(X)$ is defined on *unordered* configurations. On cohomology, this forces the normally-ordered product to be commutative: $a \cdot b = b \cdot a$. For genuinely E_1 -chiral algebras (Yangians, Etingof–Kazhdan quantum vertex algebras), commutativity fails and the PVA axiom must be replaced by a noncommutative variant. $\square$

6. THE RESOLVENT PRINCIPLE

The A_∞ operations $\{m_k\}_{k \geq 2}$ are not independent. They are determined by a single datum: the binary product m_2 .

6.1. The tree-level bootstrap equation. Let (H, p, ι, h) be a deformation retract of $(\mathcal{A}, m_1)$ onto cohomology $H = H^\bullet(\mathcal{A}, m_1)$, with $h^2 = h\iota = ph = 0$.

Definition 6.1 (Dressed propagator). The *dressed propagator* $\Phi: T^+(H) \rightarrow \mathcal{A}$ is defined by the fixed-point equation

$$\Phi = \iota + h \circ m_2 \circ \Phi^{\otimes 2}. \quad (6.1)$$

Here $T^+(H) = \bigoplus_{k \geq 1} H^{\otimes k}$ is the reduced tensor module, $\iota: H \hookrightarrow \mathcal{A}$ is the inclusion, and $h: \mathcal{A} \rightarrow \mathcal{A}$ is the contracting homotopy.

Equation (6.1) is the tree-level bootstrap: the minimal-model propagator from cohomology H into the chain algebra $\mathcal{A}$ is determined self-consistently from the free embedding ι and the interaction vertex m_2 . Iteration produces the tree formula.

6.2. The tree formula.

Theorem 6.2 (Resolvent principle). *The transferred A_∞ operations on H are given by*

$$m_k^H = \sum_{T \in \text{PRT}_k} p \circ \mathbf{m}(T) \circ \iota^{\otimes k},$$

where:

- (i) PRT_k is the set of planar rooted binary trees with k leaves;
- (ii) $|\text{PRT}_k| = C_{k-1}$, the $(k-1)$ -st Catalan number ($C_0 = 1, C_1 = 1, C_2 = 2, C_3 = 5, \dots$);
- (iii) $\mathbf{m}(T)$ assigns the binary product m_2 to each internal vertex and the contracting homotopy h to each internal edge;
- (iv) $p: \mathcal{A} \rightarrow H$ is the projection and $\iota: H \rightarrow \mathcal{A}$ is the inclusion.

Proof. The iteration of (6.1) at order k collects all ways of building a k -to-1 map from $H^{\otimes k}$ to $\mathcal{A}$ using the vertex m_2 and the propagator h . Each such composition pattern is a planar rooted binary tree with k leaves (one leaf per input from H), internal vertices labelled by m_2 , and internal edges labelled by h . The projection p extracts the cohomology class.

The number of planar rooted binary trees with k leaves is the Catalan number C_{k-1} : this follows from the recursive decomposition of a tree into its left and right subtrees, which gives the recurrence $C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}$ with $C_0 = 1$.

The Stasheff relations for $\{m_k^H\}$ are verified by checking that the boundary contributions of each tree (where two adjacent vertices merge) produce the correct composition terms. This is the combinatorial content of the associahedron: the faces of K_n correspond to the degenerate trees. $\square$

One equation, one binary input, C_{k-1} trees.

Remark 6.3 (Physical interpretation). The tree formula is the perturbative expansion of the Lagrangian self-intersection $\mathcal{L} \times_{\mathcal{M}}^h \mathcal{L}$ in a shifted symplectic stack. Each tree is a Feynman diagram with vertices m_2 and propagators h . The sum over PRT_k is the tree-level scattering amplitude at multiplicity k . Loop corrections (genus $g \geq 1$) arise from the curvature $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$, which obstructs the tree-level bootstrap at higher genus.

7. THREE COMPUTATIONS

7.1. First computation: Heisenberg (formal, class G). Let $\mathcal{A} = \mathcal{H}_k$ be the Heisenberg algebra at level k , with generator J and OPE

$$J(z)J(w) \sim \frac{k}{(z-w)^2}.$$

Proposition 7.1 (Heisenberg A_∞ structure). *The A_∞ structure of $\mathcal{H}_k$ is:*

- (i) $m_1 = 0$ (no internal differential).
- (ii) $m_2(J, J; \lambda) = k\lambda$ (the double-pole residue, absorbed by $d \log$: the OPE has a double pole, and the $d \log(z_1 - z_2)$ kernel reduces the pole order by one, producing a simple-pole collision residue k times the spectral parameter).
- (iii) $m_k = 0$ for all $k \geq 3$ (no composites, no higher poles).

The algebra is formal: the A_∞ structure is concentrated in m_2 .

Proof. The augmentation-ideal projected binary product $\bar{\mu}_2(J, J) = 0$: the OPE $J(z)J(w) \sim k/(z-w)^2$ has singular part a scalar (the identity), which projects to zero in $\bar{\mathcal{A}} = \ker(\varepsilon)$. Therefore the bar differential on $\bar{B}(\mathcal{H}_k)$ vanishes identically: all bar elements are $D_{\bar{\mathcal{A}}}$ -closed.

The operation $m_2(J, J; \lambda) = k\lambda$ is obtained from the full OPE by the $d \log$ extraction: the coefficient of $1/(z_1 - z_2)^2$ in the OPE is k , and the spectral parameter $\lambda = z_1 - z_2$ after Fourier transform gives $k\lambda$.

Since $\bar{\mu}_2 = 0$ on $\bar{\mathcal{A}}$, the degree- n bar differential involves only the identity-valued OPE residue, which lives outside $\bar{\mathcal{A}}$. All m_k for $k \geq 3$ vanish. $\square$

Corollary 7.2 (Heisenberg PVA). *On $H^\bullet(\mathcal{H}_k) = \mathbb{k}[J]$, the PVA structure is:*

- (i) Product: $J^n \cdot J^m = J^{n+m}$ (polynomial ring).
- (ii) λ -bracket: $\{J_\lambda J\} = k\lambda$.
- (iii) All higher operations $m_k^H = 0$ for $k \geq 3$.

This is a PVA of rank one. The classical r -matrix is $r(z) = k/z$.

Verification: at $k = 0$, the r -matrix vanishes ($r(z) = 0$) and the algebra reduces to a commutative ring with trivial λ -bracket: the abelian limit.

The modular characteristic is $\kappa(\mathcal{H}_k) = k$: the same scalar governs the OPE, the λ -bracket, the r -matrix, and the genus-one curvature $d_{\text{fib}}^2 = k \cdot \omega_1$.

Remark 7.3 (Shadow depth zero). The Heisenberg algebra belongs to class G in the shadow depth classification: $d_{\text{alg}} = 0$. The shadow tower terminates at $\kappa = k$; the discriminant $\Delta = 8\kappa S_4 = 0$ (since $S_4 = 0$ for the Heisenberg) confirms the finite tower. The SC-formality condition is satisfied: $\mathcal{H}_k$ is SC-formal because $m_k = 0$ for $k \geq 3$.

7.2. Second computation: Kac–Moody at level k (Lie-transverse, class L). Let $\mathcal{A} = V_k(\mathfrak{g})$ be the vacuum module of the affine Kac–Moody algebra at level $k \in \mathbb{C} \setminus \{-h^\vee\}$, with generators $\{J^a\}$ ($a = 1, \dots, d = \dim \mathfrak{g}$) and OPE

$$J^a(z)J^b(w) \sim \frac{k\kappa^{ab}}{(z-w)^2} + \frac{f^{ab}_c J^c(w)}{z-w}, \quad (7.1)$$

where κ^{ab} is the normalised Killing form and f^{ab}_c are the structure constants.

Proposition 7.4 (Kac–Moody A_∞ structure). *The A_∞ structure of $V_k(\mathfrak{g})$ is:*

- (i) $m_1 = 0$.

- (ii) $m_2(J^a, J^b; \lambda) = f^{ab}{}_c J^c + k \kappa^{ab} \lambda$. The simple-pole coefficient $f^{ab}{}_c J^c$ is the Lie bracket; the double-pole coefficient $k \kappa^{ab}$ contributes the λ -dependent term.
- (iii) $m_3 \neq 0$: the Jacobiator homotopy on $\text{FM}_3(\mathbb{C})$. The three codimension-one boundary strata of $\text{FM}_3(\mathbb{C})$ contribute $m_2(m_2(-, -), -)$ terms; their failure to cancel (the Jacobi identity defect from the simple-pole bracket) is corrected by m_3 .
- (iv) $m_k = 0$ for all $k \geq 4$, by dimension counting on $\text{FM}_k(\mathbb{C})$: the OPE has at most a double pole, so the chain-level integrands vanish for $k \geq 4$.

Proof. The binary operation extracts from the OPE (7.1): the residue at $z_1 = z_2$ with the $d \log(z_1 - z_2)$ kernel gives the simple-pole coefficient $f^{ab}{}_c J^c$ (the Lie bracket), and the double-pole coefficient $k \kappa^{ab}$ contributes the λ -linear term.

The existence of $m_3 \neq 0$ follows from the failure of strict associativity: $m_2(m_2(J^a, J^b), J^c) - m_2(J^a, m_2(J^b, J^c)) = f^{ab}{}_e f^{ec}{}_d J^d - f^{bc}{}_e f^{ae}{}_d J^d \neq 0$ (the Jacobi defect). The homotopy m_3 is the integral over $K_3 = [0, 1]$ that interpolates between the two boundary contributions.

The vanishing $m_k = 0$ for $k \geq 4$ follows from the pole structure: the OPE has at most double poles, and the weight forms η_{ij} absorb one pole order each. For $k \geq 4$ insertion points, the integrand on $\text{FM}_k(\mathbb{C})$ involves at most $(k - 1)$ weight forms but the pole excess is at most 2, which is consumed at $k = 3$. For $k \geq 4$, the integrand has no residual singularity and the FM integral vanishes. $\square$

Corollary 7.5 (Kac–Moody PVA). *On $H^\bullet(V_k(\mathfrak{g})) = \mathbb{k}[J^a : a = 1, \dots, d]$, the PVA structure is:*

- (i) *Product: polynomial ring in d generators.*
- (ii) *λ -bracket: $\{J_\lambda^a J^b\} = f^{ab}{}_c J^c + k \kappa^{ab} \lambda$.*
- (iii) *Classical r -matrix: $r(z) = k \Omega_{\text{tr}}/z$ in the trace-form convention, where $\Omega_{\text{tr}} = \sum_a J^a \otimes J_a$ is the split Casimir.*

Verification: at $k = 0$, the r -matrix vanishes ($r(z) = 0$); the algebra reduces to the commutative ring with Lie bracket $\{J_\lambda^a J^b\} = f^{ab}{}_c J^c$, which is still a well-defined PVA (the abelian limit of the level).

In the KZ convention: $r(z) = \Omega/((k + h^\vee)z)$. Bridge: $k \Omega_{\text{tr}} = \Omega/(k + h^\vee)$ at generic k .

The modular characteristic is

$$\kappa(V_k(\mathfrak{g})) = \frac{\dim(\mathfrak{g})(k + h^\vee)}{2h^\vee}.$$

At $k = 0$: $\kappa = \dim(\mathfrak{g})/2$ (not zero: the pure vacuum anomaly). At $k = -h^\vee$: $\kappa = 0$ (critical level, the bar complex is flat).

Remark 7.6 (Shadow depth one, class L). The affine Kac–Moody algebra belongs to class L: $d_{\text{alg}} = 1$. The shadow tower extends to the cubic invariant $\mathfrak{C}(\hat{\mathfrak{g}}_k)$ (from the Lie structure constants) but terminates at shadow depth $r_{\text{max}} = 3$. The quartic shadow invariant $S_4 = 0$, so the discriminant $\Delta = 8\kappa S_4 = 0$: the shadow tower is finite (class L terminates at depth 3). The algebra is *not* SC-formal: $m_3 \neq 0$.

7.3. Third computation: $\mathcal{W}_3$ (infinite tower, class M). The $\mathcal{W}_3$ algebra has two generators: the stress tensor T of conformal weight 2 and a spin-3 current W of conformal weight 3.

Proposition 7.7 ($\mathcal{W}_3$ A_∞ structure). *The A_∞ structure of $\mathcal{W}_3$ satisfies:*

- (i) $m_1 = 0$.
- (ii) $m_2(T, T; \lambda) = (c/12)\lambda^3 + 2T\lambda + \partial T$ (the full TT OPE in divided-power parametrisation; the coefficient $c/12$ matches $T_{(3)}T = c/2$ via the divided power $\lambda^{(3)} = \lambda^3/3!$).
- (iii) $m_2(T, W; \lambda) = 3W\lambda + \partial W$.

(iv) The WW operation involves the composite field $\Lambda = : (TT) : - \frac{3}{10} \partial^2 T$ at the λ -linear term:

$$m_2(W, W; \lambda) = \frac{c}{360} \lambda^5 + \frac{1}{3} T \lambda^3 + \frac{1}{2} (\partial T) \lambda^2 + \left(\frac{3}{10} \partial^2 T + \frac{32}{22 + 5c} \Lambda \right) \lambda + \dots$$

The coefficient $32/(22 + 5c)$ diverges at $c = -22/5$ (the Lee–Yang value).

(v) $m_3 \neq 0$: the Jacobiator homotopy, involving Λ .

(vi) $m_4 \neq 0$: the quartic operation from the $\text{FM}_4(\mathbb{C})$ boundary decomposition, with $32/(22 + 5c)$ entering through the $WW\Lambda$ channel.

(vii) The A_∞ structure does not terminate: $m_k \neq 0$ for all $k \geq 3$.

Proof sketch. The binary operations are extracted from the Zamolodchikov $\mathcal{W}_3$ OPE [17] using the d log kernel and divided-power parametrisation. The non-vanishing of m_3 follows from the non-Lie structure of the $\mathcal{W}_3$ bracket (the bracket involves the composite field Λ , which is not a generator). The non-vanishing of m_4 follows from the Stasheff pentagon: the m_3 - m_3 composition on ∂K_4 does not cancel (the Λ channel provides a non-trivial residue on the interior of K_4). The infinite tower $m_k \neq 0$ for all k follows by induction: each m_k generates a non-vanishing correction at degree $k + 1$ through the Stasheff cascade, driven by the non-primitive composite Λ . $\square$

Corollary 7.8 ($\mathcal{W}_3$ PVA). *On cohomology, the PVA structure is the Zamolodchikov PVA:*

$$\begin{aligned} \{T_\lambda T\} &= \frac{c}{12} \lambda^3 + 2T\lambda + \partial T, \\ \{T_\lambda W\} &= 3W\lambda + \partial W, \\ \{W_\lambda W\} &= \frac{c}{360} \lambda^5 + \frac{1}{3} T \lambda^3 + \frac{1}{2} (\partial T) \lambda^2 + \left(\frac{3}{10} \partial^2 T + \frac{32}{22 + 5c} \Lambda \right) \lambda + \dots \end{aligned}$$

The modular characteristic is $\kappa(\mathcal{W}_3) = 5c/6$: from the formula $\kappa(\mathcal{W}_N) = c(H_N - 1)$ with $H_N = \sum_{j=1}^N 1/j$, at $N = 3$ we have $H_3 = 1 + 1/2 + 1/3 = 11/6$, so $\kappa = c \cdot 5/6$.

Verification at $N = 2$: $H_2 = 3/2$, $H_2 - 1 = 1/2$, so $\kappa(\mathcal{W}_2) = c/2$, which matches the Virasoro formula $\kappa(\text{Vir}_c) = c/2$ (since $\mathcal{W}_2 = \text{Vir}$).

Remark 7.9 (Class M: infinite shadow depth). The $\mathcal{W}_3$ algebra belongs to class M: $d_{\text{alg}} = \infty$. The shadow tower $S_2, S_3, S_4, \dots$ is infinite, driven by the non-primitive composite Λ . The generating depth $d_{\text{gen}} = 3$ (the operation m_3 generates all higher m_k via the Stasheff cascade), but the algebraic depth is infinite (all $m_k \neq 0$). The discriminant $\Delta = 8\kappa S_4 \neq 0$ for $c \neq 0$ and $c \neq -22/5$, confirming the infinite tower.

This infinite depth is the algebraic shadow of the fact that three-dimensional gravity with $\mathcal{W}_3$ symmetry requires an infinite number of graviton vertices: the A_∞ structure is the perturbative expansion of the bulk theory.

Algebra	Class	d_{alg}	κ	$m_{k \geq 3}$	PVA type
$\mathcal{H}_k$	G	0	k	$= 0$	free rank-1
$V_k(\mathfrak{g})$	L	1	$\frac{d(k+h^\vee)}{2h^\vee}$	$m_3 \neq 0, m_{\geq 4} = 0$	affine, Lie-transverse
$\mathcal{W}_3$	M	∞	$5c/6$	all $\neq 0$	Zamolodchikov, infinite

8. HEPTAGON, TOPOLOGIZATION SCOPE, AND PVA DESCENT

The operad $\text{SC}^{\text{ch}, \text{top}}$ is controlled by three tests: the Vol II $\text{SC}^{\text{ch}, \text{top}}$ heptagon (seven edges routing bar, cobar, Drinfeld centre, derived centre, Swiss-cheese, PTVV, and celestial holography), the four-class topologisation scope tree, and the Khan–Zeng PVA descent for freely-generated Poisson vertex algebras.

Theorem 8.1 (The SC^{ch,top} heptagon). *The two-coloured operad SC^{ch,top} sits at the hub of a heptagon of seven equivalent presentations, each edge a named theorem:*

- (1) Operadic hub: SC^{ch,top} itself (this paper).
- (2) Koszul dual: $(\text{SC}^{\text{ch,top}})^! = (\text{Lie}^c, \text{Ass}^c, \text{shuffle-mixed})$ (this paper, Theorem 4.1; not Koszul self-dual).
- (3) Chiral factorization: the Beilinson–Drinfeld factorization algebra on $\text{Ran}(X)$ with open $\mathbb{R}$ -decoration.
- (4) BV–BRST: the Costello–Gwilliam bulk–boundary BV complex.
- (5) Convolution: the L_∞ -algebra $\text{Hom}_\alpha(C, A)$ with open-closed decoration.
- (6) Drinfeld centre: $Z(\text{Rep}_{\text{fact}}(\mathcal{A})) \simeq \text{Rep}_{\text{fact}}(Z_{\text{ch}}^{\text{der}}(\mathcal{A}))^{E_2}$ via categorified bar–cobar with half-braiding.
- (7) PTVV derived-AG: the 1-shifted symplectic mapping stack $\text{Map}(X \times \mathbb{R}_{\geq 0}, B \text{ SC-Alg})$.

Each of the seven is a distinct theorem; the Vol II heptagon chapter (sc_ckptop_heptagon.tex) proves the seven edges as named theorems.

Theorem 8.2 (Sugawara topologization by shadow class). *The passage $\text{SC}^{\text{ch,top}} \rightarrow E_3^{\text{top}}$ via Sugawara topologisation branches by shadow class:*

- Class G (Heisenberg): E_3^{top} immediate; commutative input, Dunn on F^{H_k} .
- Class $L(V_k(\mathfrak{g}), k \neq -h^\vee)$: E_3^{top} proved on the original complex, via Costello–Gwilliam factorization on $\mathbb{R} \times \mathbb{C}$ and explicit BRST primitives $\eta_1^{(i)}, \eta_1^{(ii)}$ giving $[Q, \tilde{G}_1] = T_{\text{Sug}}$ at strict chain level.
- Class $C(\beta\gamma, bc)$: E_3^{top} via Friedan–Martinec–Sbenker bosonisation reducing to class G .
- Class $M(\text{Vir}, \mathcal{W}_N)$: E_3^{top} weight-completed, via half-BRST plus MC5 pattern; original-complex chain-level is the sole remaining open direction.
- Critical level $k = -h^\vee$: E_2^{top} (a dimension drop, not a failure; Sugawara is undefined at critical level).

Without a conformal vector (or at critical level), the theory is stuck at SC^{ch,top}; this is the generic situation (Heisenberg, lattice, E_1 -chiral Yangian input, free fields without stress tensor), for which SC^{ch,top} is the final answer and E_3 the special case.

Theorem 8.3 (Khan–Zeng PVA descent on freely generated PVAs). *Let A be a freely-generated Poisson vertex algebra with conformal vector. Then the A_∞ operations $\{m_k\}$ extracted by cogenerator projection from the bar differential of the associated graded $\text{gr}_{L_i}(A)$ descend to the five PVA axioms (sesquilinearity, skew-symmetry, Jacobi, Leibniz, commutativity) on cohomology. The descent covers the 3d Poisson sigma-model class of Khan–Zeng, comprising all freely-generated PVAs with conformal vector; in particular, it recovers the Vol II orbifold route via $\mathbb{Z}/n$ -invariants preserving E_n -structure.*

Remark 8.4 (The bar complex is not an SC^{ch,top}-coalgebra). As stated in Warning 1.1, the bar complex $\overline{B}(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}})$ is an E_1 chiral coassociative coalgebra, *not* an SC^{ch,top}-coalgebra. The SC^{ch,top} structure lives on the derived centre pair $(C_{\text{ch}}^\bullet(\mathcal{A}, \mathcal{A}), \mathcal{A})$; the bar complex serves as a resolution used to compute $C_{\text{ch}}^\bullet(\mathcal{A}, \mathcal{A})$, not as an object carrying SC^{ch,top}-structure. The obstruction is formal: the bar differential is not a closed colour, deconcatenation is not an open colour, and two structures on one coalgebra do not produce the two colours of SC^{ch,top}. The SC^{ch,top} structure emerges in $\text{Hom}(B(\mathcal{A}), \mathcal{A})$, not in $B(\mathcal{A})$.

Remark 8.5 (SC^{ch,top} is Koszul, not self-dual). $(\text{SC}^{\text{ch,top}})^! \neq \text{SC}^{\text{ch,top}}$: closed-colour dimensions differ at $n \geq 3$ ($\dim \text{Com}(n) = 1$ vs $\dim \text{Lie}(n) = (n-1)!$). The duality *functor* is an involution; the operad is *not* self-dual. Livernet’s proof gives Koszulity, not self-duality.

Remark 8.6 (Status of the comparison theorems). Theorem 8.1: the seven heptagon edges are *proved* in the Vol II heptagon chapter; the operadic hub (edges 1–3) is proved directly. Theorem 8.2: class G, L, C are unconditional on the original complex; class M is weight-completed. Theorem 8.3: unconditional for freely-generated PVAs; the general class M case reduces to the weight-completed topologization.

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